Co-curricular Activities (CCA)-III: Paper Presentation Abstract

Blockchain-Based IoT Model and Experimental Platform Design in the Defence Supply Chain

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Abstract:

This article aims to design, implement, and test a blockchain-based Internet of Things (IoT) platform model for the defence industry that enables the various supply chain (SC) actors to manage transactions and information in a transparent, secure, and traceable manner, while also fostering multilateral collaboration in the supply network. The platform was designed using a private and authorized model in the SC of MBDA Italy, part of MBDA—European leader in missile industry. Blockchain technology can facilitate the decentralization and automation of SC activities. Throughput, latency, infrastructure, interoperability, onboarding, and maintenance costs, data storage costs, waste of resources, block size restrictions, and the definition of smart contracts are the primary challenges and benefits that have influenced the design of a private permissioned blockchain. Managerial implications on the necessity to integrate the supply blockchain platform into other systems (e.g., finance, insurance, and social domains) are explored to provide recommendations for the development of a more integrated digital industrial ecosystem.

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